

PROJECT 02

Custom FDM 3D Printer — Design, Build & Commissioning

Effective Engineering & Automation Pvt Ltd
Feb 2023 – Aug 2023 · Automation Engineering Intern

MUDITHA PRIYASAD

Mechanical & Automation Engineer
BSc (Hons) Mech. Eng. — Univ. of Jaffna
SOLIDWORKS CSWA · CSWA-AM · IESL
AM-32795
muditha00@icloud.com · +94 71 7599 667

PROJECT SUMMARY

Designed and built a custom FDM 3D printer from scratch for in-house use. Covered frame design, motion system, electronics, firmware configuration, and commissioning. The printer allows the company to make prototype parts and jigs the same day, at roughly half the cost of an equivalent commercial unit.

OBJECTIVES

- Build a 3D printer with a larger print area than standard desktop printers.
- Achieve print accuracy within ± 0.3 mm over 100 mm.
- Configure Marlin firmware from source to match the hardware.
- Print reliably in PLA and PETG.
- Keep total build cost at least 50% below an equivalent commercial printer.

HOW IT WORKS

Cartesian layout: the print head moves left-right (X) and front-back (Y) on GT2 belt-driven axes; the bed moves up-down (Z) on a lead screw. Four NEMA 17 steppers drive all axes. A 32-bit control board with TMC2208 drivers runs Marlin firmware compiled from source. PID-controlled hotend (max 260°C) and heated bed ensure stable printing.

KEY COMPONENTS

Component	Specification / Function
Aluminium Extrusion 20×20	V-slot frame — structural
GT2 Belt + Pulleys	X and Y axis belt drive
T8 Lead Screw (8 mm pitch)	Z axis height control
LM8UU Bearings + 8 mm Rods	Linear guides for X and Y axes
NEMA 17 Steppers ×4	X, Y, Z, and extruder drive
32-bit Control Board	Machine controller with TMC2208 drivers
All-Metal Hotend (260°C)	Melts the filament
MK3 Heated Bed, 24V	First-layer adhesion

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Motion type	Cartesian (X-Y belt, Z lead screw)	—
Print area	[TO BE FILLED]	mm W × D × H
Layer thickness	0.05 – 0.30 mm	—
Hotend max temp	260°C, PID controlled	—
Bed max temp	100°C, PID controlled	—
Firmware	Marlin 2.x (custom compiled)	Via PlatformIO
Filaments tested	PLA, PETG	—
Print accuracy	± 0.25 mm on 30 mm cube	Better than ± 0.3 mm target

MY CONTRIBUTION

- Designed full frame and moving parts in SolidWorks.
- Assembled frame, motion system, and all mechanical parts.
- Wired all electronics — board, motors, thermistors, limit switches, PSU.
- Compiled and flashed Marlin firmware from source using PlatformIO.
- Commissioned the printer: bed levelling, extruder calibration, PID tuning.
- Wrote full build and maintenance documentation.

CHALLENGES & SOLUTIONS

Frame was slightly out of square on first assembly causing uneven bed levelling — fixed by adjusting corner joints. Marlin thermal safety triggered falsely before PID tuning — resolved with M303 auto-tune. PETG moisture caused print defects — fixed by drying filament at 65°C for 4 hours.

RESULTS

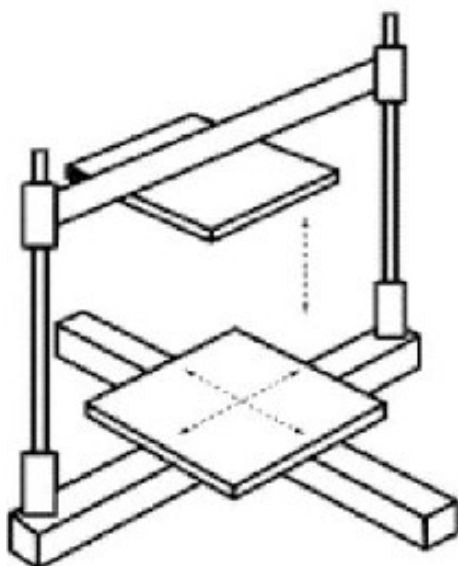
Fully working printer commissioned. Print accuracy ± 0.25 mm confirmed. Prints reliably in PLA and PETG. Build cost ~50% below equivalent commercial unit. Same-day prototype capability established for the company.

TOOLS & SOFTWARE USED

Tool / Software	Used For
SolidWorks	Frame and component design
VS Code + PlatformIO	Marlin firmware compilation and flashing
Cura	Slicer — converting models to print files
Pronterface	Printer control during commissioning
Digital Caliper	Print accuracy verification
Multimeter	Electronics wiring verification

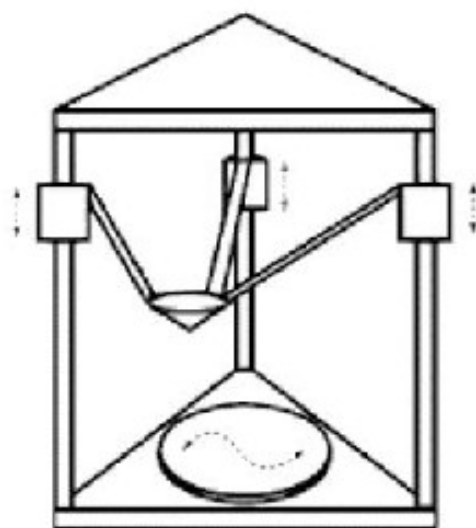
IMAGES & TECHNICAL DRAWINGS

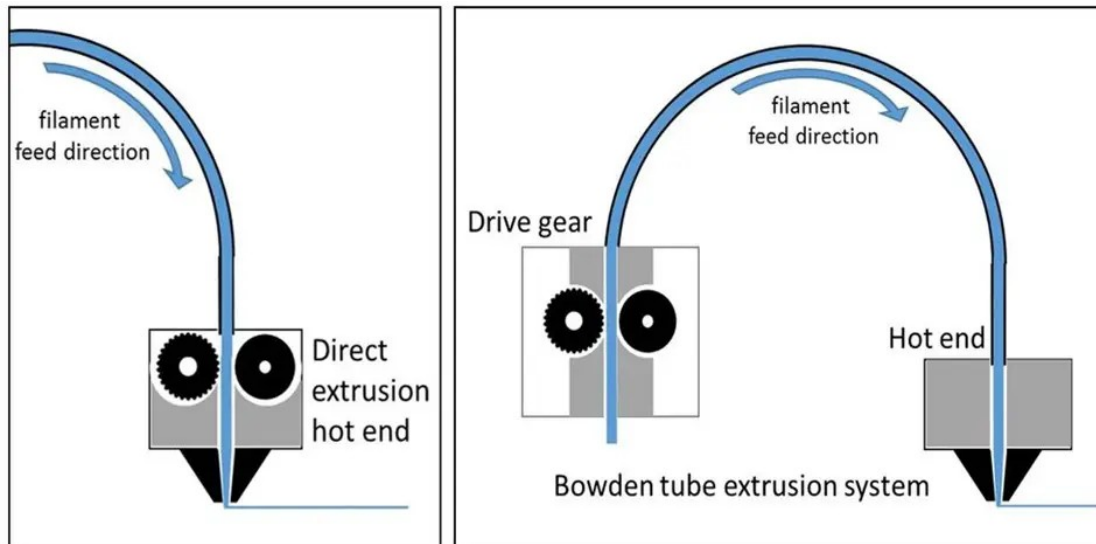
CARTESIAN



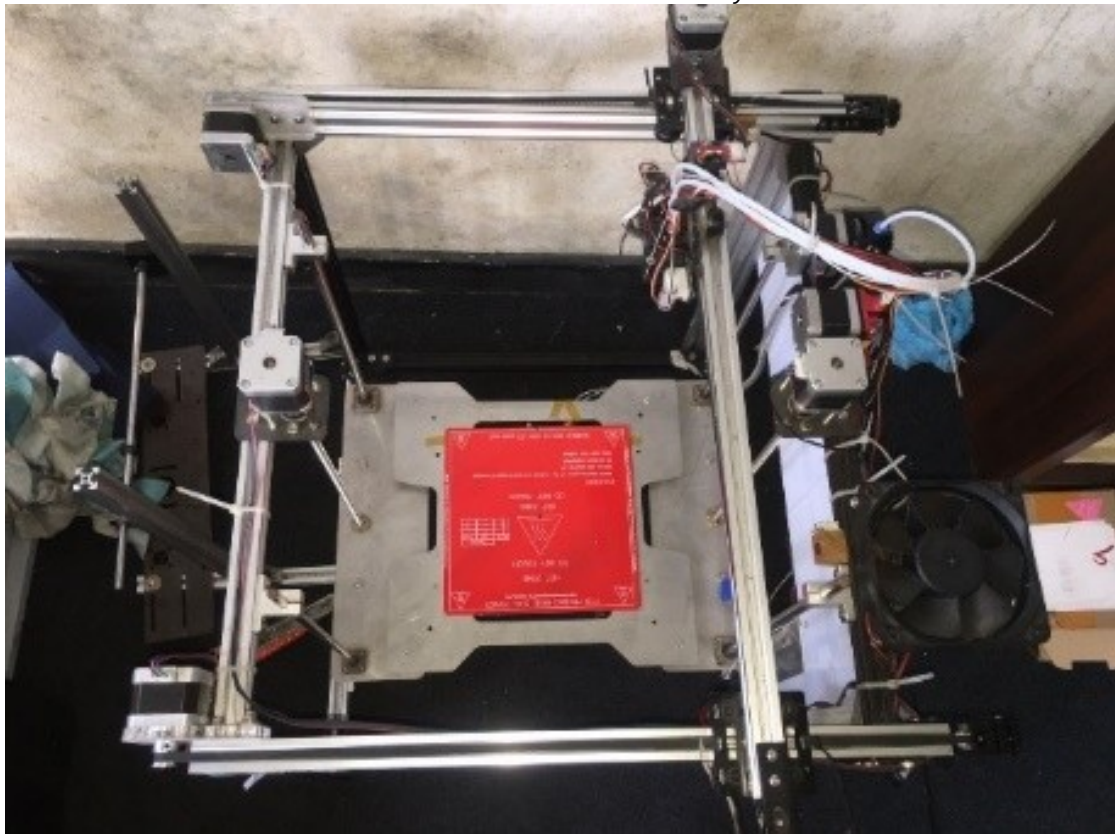
Cartesian motion system architecture

DELTA

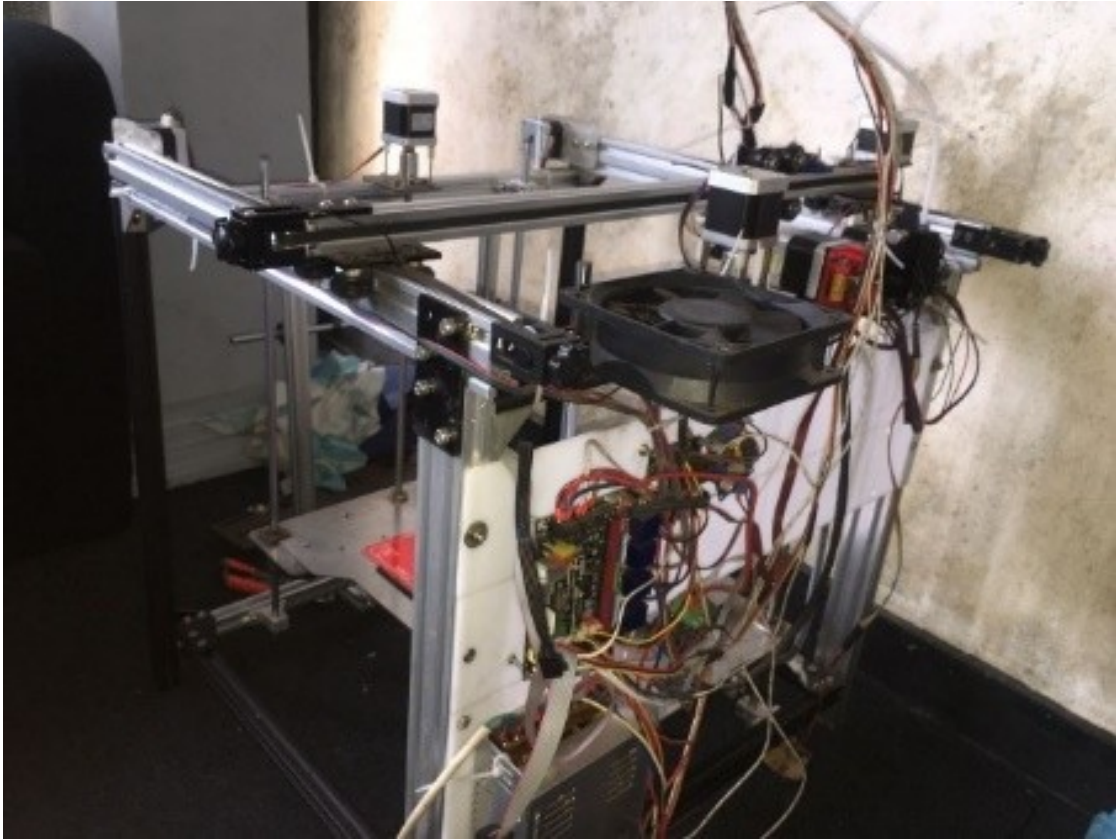




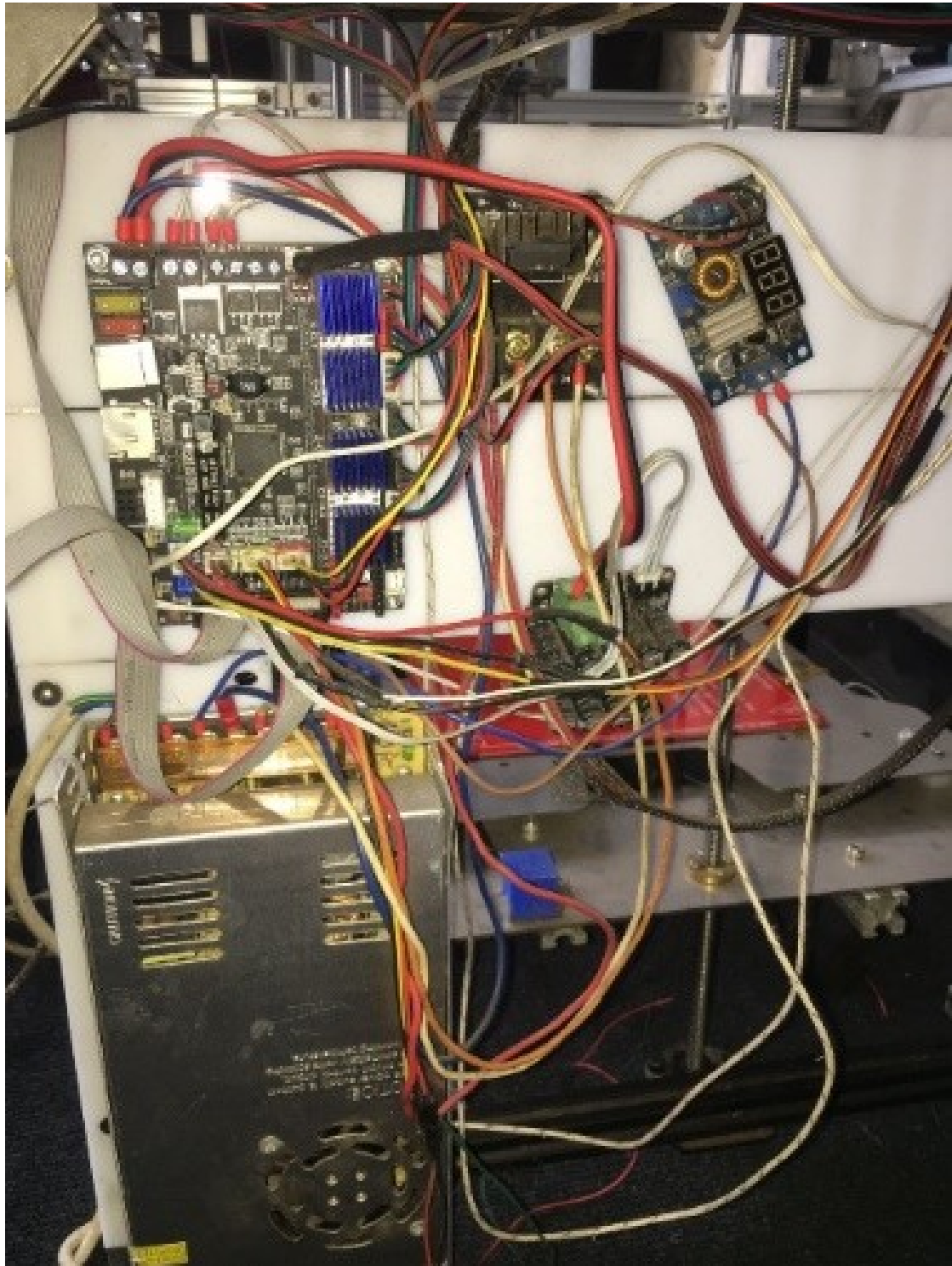
Direct-drive vs Bowden tube extrusion systems



Custom 3D printer build — overall view

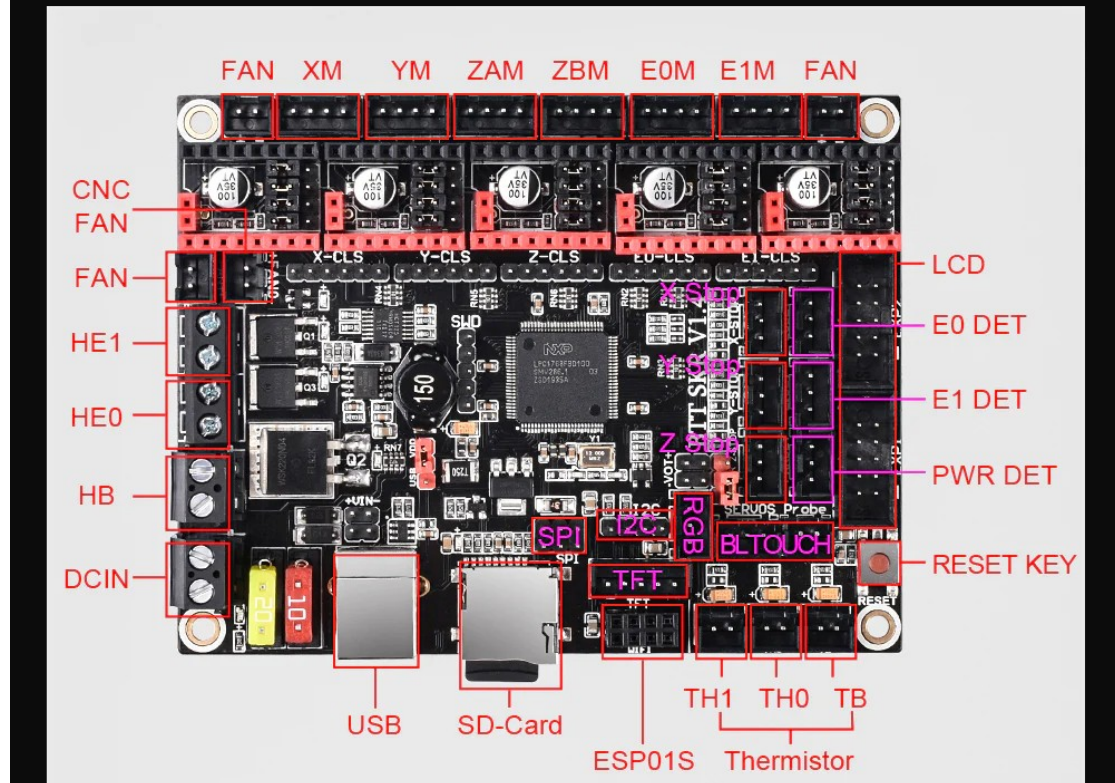


Custom 3D printer — completed assembly



Electronics control panel

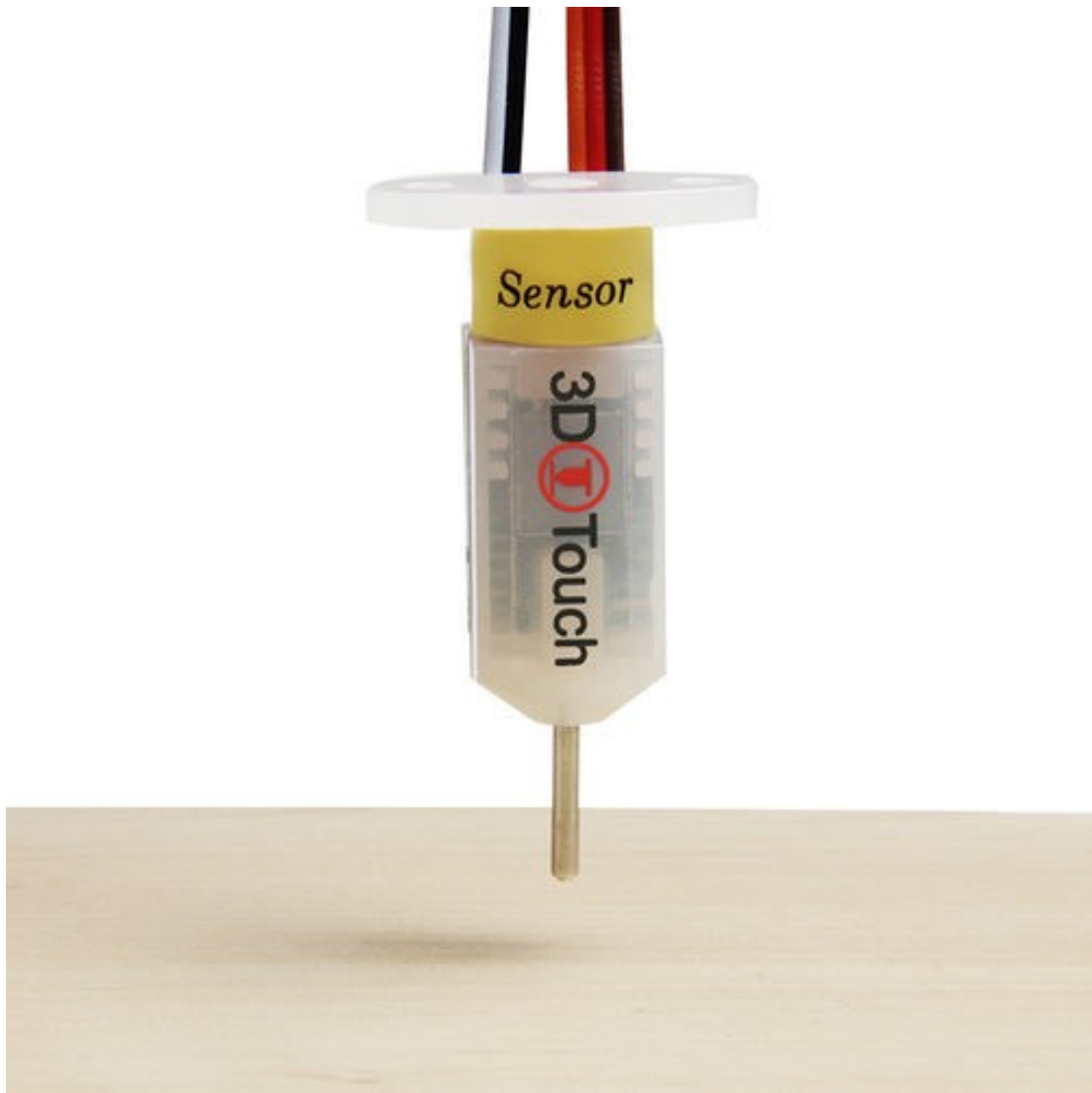
wiring diagram



BTT SKR 1.4 32-bit control board



TMC2208 stepper motor driver



BLTouch / 3D touch auto-levelling sensor